

Assignment 2

Course: Object - Oriented Programming (Code: IT069IU)

HCMIU - CSE, Spring 2026

Instructor: Le Thi Ngoc Hanh, Ph.D

1. Problem

Given a description as: A company is developing a Smart Task Reminder System that helps users receive reminders about their tasks through different reminder channels, such as email reminders, SMS reminders, and mobile app reminders. All reminder types share common information, including a reminder ID, a recipient, and a reminder message. However, each type differs in how the reminder is validated and delivered.

Every reminder must support two core operations: validating its data and sending the reminder. Although these operations are required for all reminder types, their implementations may differ depending on the delivery channel. Therefore, the system should provide a common structure for all reminders while allowing each type to define its own specific behavior.

In addition to the core operations, some reminders may support extra features such as scheduling the reminder for a future time, repeating the reminder automatically, or marking the reminder as high priority. These features are not required for all reminder types and should only be applied where appropriate.

The system must allow all reminder objects to be processed uniformly, regardless of their specific type, while still ensuring that each reminder behaves according to its own logic. Furthermore, the design should be flexible enough to support adding new reminder types in the future without modifying existing components.

2. Questions

Q1. (70 points) Based on the system description above, design a UML class diagram for the Smart Task Reminder System, showing the appropriate abstract class, subclasses, interfaces, attributes, methods, and relationships

Q2. (30 points) Explain how your design supports extensibility. If a new reminder type is added in the future, describe what changes, if any, are required in your design.

2. Guidance

- Handle this assignment by a team of 2 members.
- Your solution for this assignment MUST be prepared by handwriting.
 - The content MUST include the heading with student name, student ID, name of the assignment (*i.e.*, “Assignment 2”).
 - Submit this assignment to the teacher in-person.

IMPORTANT: You will get a penalty on total score if your submission does not follow above guidance.